A Strategy Fusion-Based Multiobjective Optimization Approach for Agile Earth Observation Satellite Scheduling Problem

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***Abstract*— Agile satellite imaging scheduling plays a vital role in improving emergency response, urban planning, national defense, and resource management. With the rise in the number of in-orbit satellites and observation windows, the need for diverse agile Earth observation satellite (AEOS) scheduling has surged. However, current research seldom addresses multiple optimization objectives, which are crucial in many engineering practices. This article tackles a multiobjective AEOS scheduling problem (MOAEOSSP) that aims to optimize total observation task profit, satellite energy consumption, and load balancing. To address this intricate problem, we propose a strategy-fused multiobjective dung beetle optimization (SFMODBO) algorithm. This novel algorithm harnesses the position update characteristics of various dung beetle populations and integrates multiple high-adaptability strategies. Consequently, it strikes a better balance between global search capability and local exploitation accuracy, making it more effective at exploring the solution space and avoiding local optima. The SFMODBO algorithm enhances global search capabilities through diverse strategies, ensuring thorough coverage of the search space. Simultaneously, it significantly improves local optimization precision by fine-tuning solutions in promising regions. This dual approach enables more robust and efficient problem-solving. Simulation experiments confirm the effectiveness and efficiency of the SFMODBO algorithm. Results indicate that it significantly outperforms competitors across multiple metrics, achieving superior scheduling schemes. In addition to these enhanced metrics, the proposed algorithm also exhibits advantages in computation time and resource utilization. This not only demonstrates the algorithm's robustness but also underscores its efficiency and speed in solving the MOAEOSSP.**

***Index Terms*— Agile Earth observation satellite (AEOS), multiobjective dung beetle optimization (MODBO), remote sensing, satellite observation scheduling.**

NOMENCLATURE

Sat	$\text{Sat} = \{\text{sat}_1, \text{sat}_2, \ldots, \text{sat}_i, \ldots, \text{sat}_n\}$, Sat represents the set of satellites, and n denotes the number of satellites.
T	$T = \{t_1, t_2, \ldots, t_j, \ldots, t_m\}$, T represents the set of observation tasks, and m denotes the number of tasks.
O^i	$O^i = \{o_1^i, o_2^i, \ldots, o_k^i, \ldots, o_q^i\}$, O^i represents the orbit set of the ith satellite, and q denotes the number of orbits.
Vtw	$\text{Vtw} = \{\text{vtw}_1, \text{vtw}_2, \ldots, \text{vtw}_l, \ldots\}$, Vtw represents the set of visible time windows, r represents the number of the windows $\text{vtw}_r = \{st_r, et_r, dt_r\}$, vtw_r represents the rth visible time window, st_r represents the visible start time, et_r represents the visible end time, and dt_r represents the visible duration.
Otw	$\text{Otw} = \{\text{otw}_1, \text{otw}_2, \ldots, \text{otw}_r, \ldots\}$, Otw represents the set of observable time windows, r represents the number of the windows, $\text{otw}_r = \{bt_r, ft_r, ct_r\}$, otw_r represents the rth observable time window, bt_r represents the observable begin time, ft_r represents the observable finalization time, and ct_r represents the observable continuance.
${A_t}_j^h$	${A_t}_j^h$ represents the attitude transfer time of the jth task to the hth task.
E	$E = \{e_1, e_2, \ldots, e_n\}$, E represents the energy set of the satellites and n represents the number of satellites.
E_i^j	E_i^j represents the energy consumed by the ith satellite for the jth task.
Et_i	Et_i represents the energy consumption per unit time of the attitude maneuver of the ith satellite during task transition.
Mem	$\text{Mem} = \{\text{mem}_1, \text{mem}_2, \ldots, \text{mem}_n\}$, mem the set of memory capacity of the satellite, and n represents the number of satellites.
M_i^j	M_i^j represents the memory space consumed by the ith satellite observing the jth task.
P	$P = \{p_1, p_2, \ldots, p_m\}$, where Prio represents the set of tsak priorities, m represents the number of tasks.
x	x represents a bool variable, the observation task is 1, otherwise it is 0.

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$(y_j)_i^k$ $(y_j^i)^k$ represents a bool variable, taking the value 1, the jth task of the ith satellite in the kth orbit is observed, and 0 otherwise.

$(z_j^h)_i^k$ $(z_j^h)_i^k$ represents a bool variable, taking the value 1, the hth task is the successor of the jth task on the kth orbit of the ith satellite, and 0 otherwise.

I. Introduction

AGILE Earth observation satellites (AEOSs) possess the capability for rapid attitude maneuvering around any Euler axis, offering quick stabilization, rapid deployment, and multitarget observation. These advantages significantly enhance the capability and efficiency of target observation. By utilizing optical cameras for imaging the Earth's surface, agile satellites can provide users with the necessary remote sensing image products. Due to their superior imaging quality, extensive observation range, and the advantage of not being restricted by national air and sea space, agile satellites can operate stably over long periods. Consequently, they are widely applied in meteorological forecasting, environmental protection, surface mapping, and disaster monitoring [1], [2].

The AEOS scheduling problem (AEOSSP) involves allocating observational time windows to targets based on the current available resources of the satellite, while adhering to operational constraints to create an optimal satellite scheduling plan [3]. As illustrated in Fig. 1, the observable region is determined by the satellite's altitude, sensor field of view, and axial angle. Due to the increased flexibility in observation offered by AEOS, these satellites allow for comparatively longer time windows for tasks, enabling the assignment of tasks to any available time slots of one or more AEOS. However, this process involves not only selecting which tasks to execute but also determining the most appropriate time windows, making AEOSSP a highly complex, NP-hard problem, as noted in [4].

As the number of agile satellites continues to grow due to decreasing launch costs, and the complexity and diversity of user demands continue to expand, existing approaches to solving AEOSSP are limited in their ability to address these challenges. Many current methods focus on optimizing specific aspects, such as task selection or time window allocation, but often neglect the multiobjective nature of the problem, which requires balancing competing factors like observational profit, energy consumption, and load balancing. This gap highlights the need for more advanced and scalable optimization techniques that can better accommodate the increasing number of observation windows and the evolving demands of users. Thus, the study of AEOSSP and the development of novel solutions, and methods are not only of practical importance but also critical for advancing theoretical knowledge in satellite scheduling [5], [6].

Algorithms for solving the AEOSSP can be broadly classified into exact, heuristic, and intelligent optimization algorithms. The exact algorithms use mathematical optimization algorithms [7], [8], such as branch-and-price [9], branch-cut-and-price [10], and dynamic programming [11], search for optimal solutions. These methods are often enhanced with column generation [9], Lagrangian relaxation [12], and other techniques to reduce complexity and improve efficiency. However, due to the AEOSSP's complexity, exact algorithms can be computationally intensive and may fail to find solutions within a reasonable timeframe. Heuristic algorithms rely on experience and intuition to quickly find feasible solutions. While they do not guarantee globally optimal solutions, they often produce high-quality results within a reasonable time. Examples include greedy algorithms [4] and local search techniques [4], [13]. Intelligent optimization algorithms, inspired by natural biological or physical phenomena, use mechanisms such as evolution, cooperation, and competition for global search. These are particularly suited for complex optimization problems. Examples include genetic algorithms [14], ant colony optimization (ACO) [15], and particle swarm optimization [16]. Consequently, most current research focuses on intelligent optimization algorithms for solving the AEOSSP.

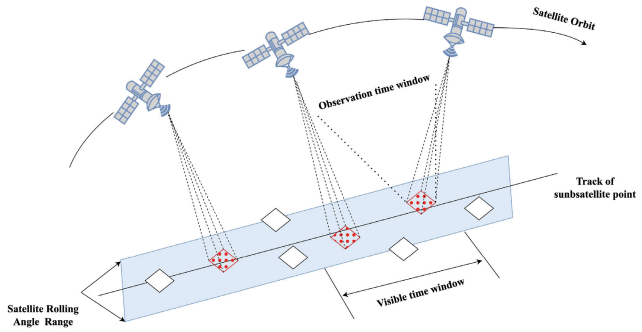


Fig. 1. Schematic of the working of AEOS.

Extensive research on the AEOSSP has revealed that it typically requires numerous ground observations to accomplish target missions and calculate the observational benefits of satellite scheduling. Therefore, the primary objective of most studies is to maximize these observational benefits.

Here are some classic models and methods for maximizing the observational returns of the AEOSSP objective function. For example, Xu et al. [17] established a mathematical model and introduced a genetic algorithm. Through three phases—discretization, target decomposition, and scheduling—they successfully generated the optimal observation plan, maximizing total profit. The effectiveness and reliability of this framework were validated through actual computational experiments. Peng et al. [18] proposed a scheduling model and method based on time-correlated transition times. By considering satellite resource constraints and visibility constraints, they generated scheduling input data, established a scheduling model with multiorbit and transition time constraints, and used first-in–first-out (FIFO) rules and the triangular inequality for preprocessing. The success of this method in effectively arranging satellite observation tasks was ultimately validated. He et al. [19] modeled the AEOSSP as a Markov decision process (MDP) and applied reinforcement learning methods to solve task sequencing and timing problems, achieving efficient satellite task scheduling and maximizing total profit. Zhou et al. [15] addressed the variable-duration imaging problem for AEOS by first applying a diminishing marginal

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returns function. They proposed an improved adaptive ACO (IAACO) algorithm. Computational experiments using actual AEOS parameters demonstrated that IAACO outperformed traditional ACO and adaptive large neighborhood search (A-ALNS) [20] algorithms in terms of benefits and convergence. Chu et al. [21] designed an AEOSSP model based on a branch-and-bound algorithm. This model utilizes the concepts of "sequence solution" and "schedule solution" and achieves optimal task arrangement by employing a greedy schedule constructor, symmetry-breaking, and subspace pruning strategies, thereby maximizing observational benefits while satisfying time window and maneuver constraints. He et al. [20] proposed an A-ALNS algorithm for the scheduling of multiple agile satellites. This algorithm optimizes scheduling through adaptive task allocation and various heuristic methods. Experimental results show that A-ALNS outperforms existing methods in terms of task completion and robustness.

In the AEOSSP, the transmission of observational data plays a crucial role. Researchers have conducted a series of studies on data-driven methods to enhance observational efficiency and data download profits. Du et al. [22] developed a data-driven parallel scheduling method, employing the cooperative neuro-evolution of augmenting topologies (C-NEATs) probability prediction model and data-driven task allocation strategies. Combined with a genetic algorithm and local search, this parallel scheduling algorithm optimizes the scheduling of multiple AEOS, maximizing observational efficiency. Wu et al. [23] proposed a frequent pattern-based parallel search algorithm (FPBPS), which integrates parallel local search and frequent pattern mining techniques to optimize AEOS scheduling. Experiments demonstrate that FPBPS outperforms existing algorithms across different task scales. He et al. [24] introduced a knowledge reasoning approach, formulating scheduling strategies based on the coupling relationships of satellite subsystems. By optimizing objective functions and multiple constraint conditions, this method maximizes observational and data download profits, improving scheduling adaptability and computational efficiency. Kandepi et al. [25] proposed a heuristic search strategy based on greedy algorithms and dynamic programming to address the scheduling of AEOS constellations for large-area targets. Experimental validation confirms that this strategy outperforms traditional methods in both performance and applicability.

To address the challenge of cloud cover uncertainty in Earth observation, Wang et al. [26] proposed a novel mathematical model and designed both exact and heuristic algorithms. By adopting a scheduling strategy that assigns each task to multiple orbits and combining it with an expectation model, they increased the probability of successful task observation, demonstrating the superiority of the proposed solution. In addition, Wang et al. [27] proposed an expectation model and a sample average approximation (SAA) model. By effectively solving the expectation model through a branch-and-price algorithm, they demonstrated that the SAA model's solutions are more robust. However, these methods are geared toward nonagile satellites and have certain limitations when applied to agile satellites. Wang et al. [28] introduced the concept of a budget uncertainty set, established a linearized robust scheduling model, and proposed a hybrid heuristic algorithm that combines column generation and simulated annealing. Extensive simulation experiments validated the effectiveness and efficiency of this method. Han et al. [29], using a heuristic method based on simulated annealing, solved the scheduling problem for multiple agile satellites with a quick insertion algorithm to maximize observational benefits. Experimental results showed that this method performed well across different scheduling instances, finding relatively optimal solutions within a reasonable timeframe.

In the aforementioned research, various methods have been proposed to address the AEOSSP with a single objective function or by weighting multiple objective functions into a single objective function. These studies have achieved good results, but in practical engineering applications, it is often necessary to meet the diverse needs of different users. In recent years, research on multiobjective AEOSSP (MOAEOSSP) problems has gradually unfolded. For MOAEOSSP, multiobjective evolutionary algorithms (MOEAs) have become an effective method for solving multiobjective optimization problems in engineering practice [30]. MOEAs are generally classified into three categories: Pareto-dominance-based [31], decomposition-based [32], and indicator-based approaches [33], [34]. Despite the development of numerous MOEAs in the literature to solve similar problems, for example, in the multiobjective satellite range scheduling problem, Du et al. [35] established a decision model and employed a MOEA along with a local search algorithm to maximize the execution rate of communication requests and optimize antenna utilization. This approach provided high-quality scheduling schemes, helping satellite monitoring departments formulate reasonable plans, extend antenna life, and enhance emergency response capabilities. Song et al. [36] proposed a learning-guided nondominated sorting genetic algorithm II (LGNSGAII) and task-time window selection algorithm (TTSA), optimizing the total profit of task sequences and minimizing the failure rate. They verified the feasibility and efficiency of the algorithm, successfully addressing the optimization of execution order and execution time for satellite telemetry and command tasks. Wang et al. [37] proposed a MOEA combining problem-specific design and a diversity-convergence balance mechanism, experimentally verifying its superior performance in convergence and diversity.

However, this multiobjective prospective has not yet considered the optimization and tradeoffs of the energy consumption dimension as discussed in this article. First, most multiobjective satellite scheduling methods focus on the satellite range scheduling problem, which has different requirements and constraints compared to MOAEOSSP. In MOAEOSSP, besides considering the visible time window, it is also necessary to consider satellite resource consumption to ensure higher gains with lower energy consumption. Second, although there are a few studies on MOAEOSSP, they usually only consider bi-objective optimization. For example, the objective functions considered in the literature include maximizing the schedule profits of all selected tasks and maximizing the schedule

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slack time [3]; minimizing the failure rate and optimizing load balance [38]; improving the average imaging resolution and reducing energy consumption [39]; and maximizing the total profit while enhancing the average image quality of scheduled requests [40]. However, these papers do not address the crucial aspect of satellite load balancing. Our study sets three optimization objectives from the perspectives of tasks and resources: maximizing observation gains, minimizing energy consumption, and achieving satellite load balancing. Compared to bi-objective optimization, solving these three optimization objectives entails a larger search space, which is crucial for enhancing solution diversity, distribution uniformity, and global search capability. This complexity makes our research more challenging and necessitates novel, innovative solution methods.

The main contributions of this article can be summarized as follows.

1) We established a novel multiobjective optimization mathematical model tailored to the specific characteristics and practical engineering requirements of MOAEOSSP. This model simultaneously addresses three optimization objectives that are rarely considered in previous research: maximizing observational profit, minimizing energy consumption, and achieving load balancing.
2) To address the challenges posed by the increased search space and complexity of three objective functions, we propose a novel strategy-fused multiobjective dung beetle optimization (SFMODBO) algorithm. This algorithm leverages the position update characteristics of different dung beetle populations, combined with highly adaptive strategies, to effectively balance global search capability and local exploitation precision.
3) The effectiveness and applicability of SFMODBO have been validated through extensive simulation experiments. Compared with three classical MOEAs, strength Pareto evolutionary algorithm 2 (SPEA2), MOEA based on decomposition (MOEA/D), MOPSO, and the standard multiobjective dung beetle optimization (MODBO) algorithm, the proposed algorithm demonstrates superior performance across multiple evaluation metrics.

The structure of this article is organized as follows. Section II provides a detailed description and mathematical model of MOAEOSSP. Section III introduces the proposed MOEA. Section IV presents the simulation experiments and analyzes the results. Finally, Section V summarizes the main findings of this study and discusses potential directions for future research.

II. MOAEOSSP Model

The satellite scheduling process typically aims to select and arrange user-submitted observation requests within given constraints. These requests involve acquiring images of specific areas on Earth, with task priorities predetermined by users or decision-makers. Most images are stored on onboard devices, while some are immediately transmitted to ground stations or relay satellites. In addition, AEOS requires attitude adjustments between two tasks, with the duration dependent on the start times and observation durations of the preceding and subsequent tasks. Since observation requests often exceed AEOS's capabilities, constrained by onboard storage and energy limitations, this issue is categorized as an oversubscription problem.

Unlike previous studies [38], [40] that employed dual-objective functions, this article introduces three new objective functions: maximizing observation revenue, minimizing energy consumption, and achieving load balancing. First, maximizing observation revenue ensures the overall effectiveness of satellite missions, thereby enhancing the efficiency of observation tasks and user satisfaction. Second, minimizing energy consumption aims to optimize resource utilization, which is crucial for extending the satellite's operational lifespan and ensuring long-term stability. Finally, achieving load balancing ensures a reasonable distribution of tasks in terms of time and space, preventing excessive concentration or idleness of resources, thereby improving the overall efficiency and fairness of task scheduling.

A. Assumptions

Since MOAEOSSP is a highly complex scheduling problem with three optimization objectives and multiple constraints, it is necessary to make the following reasonable assumptions and simplifications.

1) *We Consider Only Target Points Observable in a Single Pass:* This makes scheduling easier and reduces complexity. It also matches real satellite operations, as a satellite can only observe targets in its view during one pass.
2) *Each Observation Task Cannot be Preempted, and a Satellite Can Only Observe One Task at a Time:* The nonpreemptibility of observation tasks and the single-task constraint simplify the scheduling model. This reflects real operational limitations, as switching tasks can result in data loss or reduced efficiency. This assumption also avoids task conflicts and enhances scheduling feasibility.
3) *We Ignore Data Downlink and Recharging Processes:* This reduces the model's complexity, making the scheduling problem more manageable. While important, these processes are considered secondary factors in the initial optimization model, allowing focus on the core scheduling problem.
4) *We do Not Consider Satellite Malfunctions:* Not considering malfunctions simplifies the scheduling model, as introducing them would significantly increase the problem's complexity. Assuming normal satellite operations throughout the scheduling period makes the model more manageable and optimizable.

B. Constraints and Objective Function

The AEOSSP involves the efficient management of multiple satellites and their respective observation tasks, aiming to optimize resource utilization and maximize task completion.

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To achieve this goal, we define a series of symbols and variables to establish a clear model that describes the relationship between satellites and tasks. For instance, Sat represents the set of satellites, T represents the set of tasks, O^i represents the set of orbits for the ith satellite, and Vtw and Otw represent the sets of visible and observable time windows, respectively. These symbol definitions help to clearly describe the interrelationships and constraints between satellites and tasks during the modeling process.

In terms of specific methods and steps, we first define each variable and its calculation methods, such as the attitude transition time ${A_t}_j^h$, energy consumption E_i^j, and storage space M_i^j. These definitions lay the foundation for further optimization. Next, we use Boolean variables x_j and $(y_j)_i^k$ to represent task execution status, where x_j indicates whether a task is executed, and $(y_j)_i^k$ indicates whether the ith satellite executes the jth task on the kth orbit. Another Boolean variable, $(z_j^h)_i^k$, indicates whether the hth task is the successor of the jth task on the kth orbit of the ith satellite; it takes the value of 1 if true and 0 otherwise. Detailed definitions are provided in Nomenclature. Based on the aforementioned descriptions and the detailed definitions in Nomenclature, the MOAEOOSP is formally defined by the following constraints and objective functions.

1) *Objective Functions:*

$$\min F(X) = \{f_1(X), f_2(X), f_3(X)\} \tag{1}$$

$$f_1 = \sum_{i=1}^{n}\sum_{k=1}^{q}\sum_{j=1}^{m} p_j \cdot (y_j)_i^k \tag{2}$$

$$f_2 = \sum_{i=1}^{n}\sum_{k=1}^{q}\sum_{j=1}^{m}\left(E_i^j \cdot (y_j)_i^k + \sum_{h=1,h\neq j}^{m} \left(z_j^h\right)_i^k \cdot {A_t}_j^h \cdot Et_i\right) \tag{3}$$

$$f_3 = \frac{1}{n}\sum_{i=1}^{n}(E_i - \overline{E})^2 \tag{4}$$

where E_i and $\overline{E}$ of f_3 are

$$E_i = \sum_{k=1}^{q}\sum_{j=1}^{m}\left(E_i^j \cdot (y_j)_i^k + \sum_{h=1,h\neq j}^{m} {A_t}_j^h \cdot \left(z_j^h\right)_i^k \cdot Et_i\right)$$

$$\overline{E} = \frac{1}{n}\sum_{i=1}^{n} E_i. \tag{5}$$

2) *Constraints:*

$$\forall t_j \in T : \sum_{i=1}^{n}\sum_{k=1}^{q}\sum_{j=1}^{m}(y_j)_i^k \leq 1 \tag{6}$$

$$\forall\, r_1, r_2 \in \{0, 1, 2, \ldots, N_1\}, r_1 \neq r_2, l \in \{1, 2, \ldots, N_2\}$$
$$\exists\, owt_{r1,r2} \in \text{Otw}, \text{vtw}_l \in \text{Vtw} : x_r(bt_r, ft_r) \subseteq x_l(st_r, et_r)\ \&\ ft_{r1} \leq bt_{r2} \tag{7}$$

$$\forall\, t_j, t_h \in T, \quad j \neq h, \ \left(z_j^h\right)_i^k = 1 : bt_j + ct_j + {A_t}_j^h \leq bt_h \tag{8}$$

$$\forall i \in 1, 2, \ldots, n, \quad k \in 1, 2, \ldots, q, j, h \in 1, 2, \ldots, m \quad j \neq h, \quad (y_j)_i^k = \left(z_j^h\right)_i^k = 1 : \sum_{k=1}^{q}\sum_{j=1}^{m}\left(E_i^j \cdot (y_j)_i^k + \sum_{h=1,h\neq j}^{m} {A_t}_j^h \cdot \left(z_j^h\right)_i^k \cdot Et_i\right) \leq e_i \sum_{k=1}^{q}\sum_{j=1}^{m} M_i^j \cdot (y_j)_i^k \leq \text{mem}_i. \tag{9}$$

The three objective functions we propose are (2)–(4). These objective functions aim not only to maximize the total observation profit and minimize energy consumption but also to achieve load balancing among the satellites. The metric for load balancing is to minimize the variance in energy consumption among the satellites. Constraint (6) ensures that each task is observed only once, preventing redundant observations. Constraint (7) specifies that the observation time for each task must fall within its visibility window and not overlap with other tasks. Constraint (8) guarantees sufficient transition time between tasks for the satellite to perform the required attitude maneuvers. Finally, constraint (9) ensures that the satellite's available resources, such as energy and capacity, are not exceeded during scheduling.

III. SFMODBO Algorithm

Currently, there are various algorithms available for solving multiobjective optimization problems, such as SPEA2 [41] and MOEA/D [32]. However, these algorithms exhibit significant drawbacks, such as easily getting trapped in local optima and being computationally heavy. To address these issues, we propose a novel MODBO algorithm. By integrating various strategies based on different dung beetle population characteristics, this algorithm significantly enhances global and local search capabilities. The dung beetle optimization (DBO) algorithm is a population-based optimization method inspired by the behaviors of dung beetles, including ball-rolling, dancing, foraging, stealing, and breeding. Its objective is to achieve fast convergence and high-precision solutions [42]. MODBO employs the popular and effective multiobjective optimization technique known as nondominated sorting to enhance its performance across various domains [43]. Specifically, this algorithm has demonstrated excellent performance in port berth allocation, energy scheduling [44], and wind-thermal coordinated load frequency control systems [45], verifying its feasibility. However, due to the higher complexity of the multiobjective algorithm for MOAEOSSP compared to the other two fields, existing methods cannot be directly applied.

A. MODBO Algorithm Based on Multistrategy Fusion

According to the DBO algorithm [42], dung beetles are categorized into four types: ball-rolling dung beetles, breeding dung beetles, small dung beetles, and thieving dung beetles. Their proportions are 4:4:5:7, respectively, as illustrated in Fig. 2. To effectively enhance the overall performance of the

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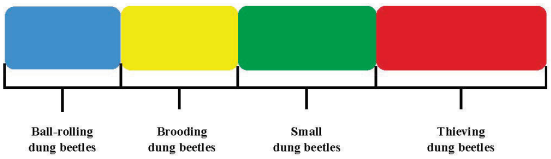


Fig. 2. Proportional distribution of the four dung beetles.

algorithm, avoid local optima, and address the complexities discussed in this study, it is necessary to modify the algorithm. Therefore, we propose an improved mathematical model to better tackle these challenges. The improved mathematical model is as follows.

1) *Rolling Ball:* The rolling behavior of dung beetles affects the global search ability and convergence speed of algorithms. However, the standard DBO algorithm's linear rolling limits local search and easily falls into local optima. To address this, the golden sine algorithm (Golden-SA) [46] is introduced. It traverses sine function values to cover the unit circle and uses the golden ratio to narrow the solution space, achieving a balance between exploration and exploitation. The advantage of the golden sine algorithm lies in its ability to traverse sine values and use the golden ratio, which effectively covers the unit circle and narrows the solution space, providing a balanced approach to global exploration and local exploitation. In contrast, linear rolling strategies often underperform when searching complex, multiextremum problems, such as the Rastrigin function, which is characterized by a large number of local minima. Linear rolling tends to get stuck in suboptimal solutions, while the golden sine strategy more effectively avoids this by enhancing global search capabilities and maintaining precision in local searches. To more clearly illustrate the effects of incorporating the golden sine approach, Fig. 3 highlights the impact of both the linear rolling and Golden-SAs on search capabilities, particularly when tested on the Rastrigin function. This enhances both global and local search capabilities. Therefore, the position update for the rolling behavior is defined as follows:

$$x_i(t+1) = |\sin\theta_1|x_i(t) - \theta_2 \sin\theta_1|c_1 X^W - c_2 x_i(t)| \tag{10}$$

where t indicates the current iteration number, $x_i(t)$ indicates the position of the ith iteration dung beetle, θ_1 is a random number in the range $[0, 2\pi]$ that determines the movement distance, θ_2 is a random number in the range $[0, \pi]$ that determines the direction of the position update, c_1 and c_2 are the golden ratio coefficients, and X^W indicates the global worst position.

To prevent linear decrement from causing rapid convergence and trapping in local optima, a nonlinear weight w is introduced. This weight gradually decreases as the number of iterations increases, ensuring strong global search capability in the early stages while slowing down the convergence speed in the later stages. With the introduction of adaptive weight, the position update is as follows:

$$\begin{aligned} x_i(t+1) &= (1-w)|\sin(\theta_1)|x_i(t) \\ &\quad -w\theta_2 \sin(\theta_1)|c_1 X^W - c_2 x_i(t)| \\ w &= w_{\min} + (w_{\max} - w_{\min}) \cdot e^{-\alpha \cdot \frac{t}{T_{\max}}} \end{aligned} \tag{11}$$

where $w_{\max}$ is the maximum weight, $w_{\min}$ is the minimum weight, α is the attenuation factor, and $T_{\max}$ is the maximum number of iterations.

2) *Breeding:* Dung beetles roll dung balls to a safe place and hide them to choose a secure site for laying eggs, as shown in Fig. 4. Based on this, a boundary selection strategy is proposed to simulate the egg-laying area of female dung beetles, described as follows:

$$\begin{aligned} \text{Lb}^* &= \max(X^* \times (1-R), \text{Lb}) \\ \text{Ub}^* &= \min(X^* \times (1+R), \text{Ub}) \end{aligned} \tag{12}$$

where X^* is the current local best position, Lb^* and Ub^* are the lower and upper bounds of the spawning area, $R = 1 - (t/T_{\max})$ with $T_{\max}$ as the maximum iteration number, and Lb and Ub are the optimization problem's lower and upper bounds. The female dung beetles select a hatching ball to lay eggs, with each female dung beetle laying only one egg per iteration. The boundaries of the egg-laying area and the position of the breeding ball dynamically change with the R-value. Therefore, the position information of the breeding ball is as follows:

$$\begin{aligned} B_i(t+1) &= X^* + b_1 \times (B_i(t) - \text{Lb}^*) \\ &\quad + b_2 \times (B_i(t) - \text{Ub}^*) \end{aligned} \tag{13}$$

where $B_i(t)$ represents the position of the ith brood ball at the tth iteration, b_1 and b_2 are two independent random vectors of size $1 \times D$, and D indicates the dimension of the optimization problem.

3) *Foraging:* When the eggs hatch into young dung beetles, an optimal foraging area is established to guide them in simulating their natural foraging process. The definition is as follows:

$$\begin{aligned} \text{Lb}^b &= \max(X^b \times (1-R), \text{Lb}) \\ \text{Ub}^b &= \min(X^b \times (1+R), \text{Ub}) \end{aligned} \tag{14}$$

where X^b represents the global best position, and Lb^b and Ub^b are the lower and upper bounds of the optimal foraging area, respectively. Thus, the position of the small dung beetle is updated as follows:

$$\begin{aligned} x_i(t+1) &= x_i(t) + \text{Cauchy}((x_i(t) - \text{Lb}^b) \\ &\quad + (x_i(t) - \text{Ub}^b)) \end{aligned} \tag{15}$$

where $x_i(t)$ represents the position of the ith small dung beetle at the tth iteration; Cauchy is a 1-D Cauchy mutation function, it is a probability distribution with heavy tails, which allows it to generate larger step sizes and thus help the algorithm escape local optima.

4) *Thefting:* The number of theft dung beetles in a population typically exceeds one-third, which helps achieve rapid convergence of the population. However, standard algorithms tend to focus on perturbations near the

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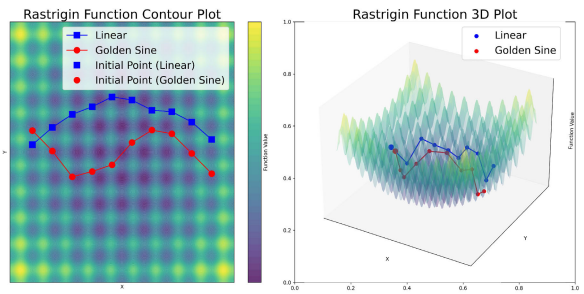


Fig. 3. Illustration of local optimum search capabilities: linear rolling versus golden sine method.

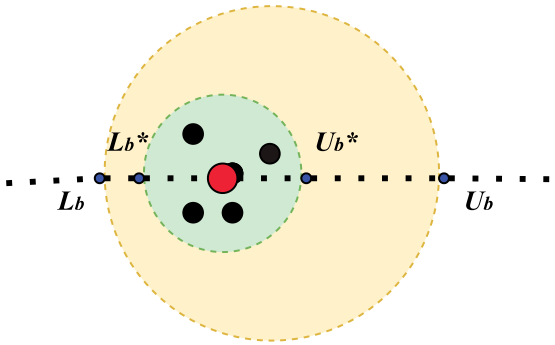


Fig. 4. Conceptual model of the boundary selection strategy.

global optimal position, failing to fully utilize current position information. This results in insufficient global exploration capability of the population and a tendency to converge prematurely when solving load problems. Therefore, at the early stages of iteration, it is desired to fully exploit the current position information, starting from the current position and progressing with large steps guided by both the global optimal position and the local optimal position. In subsequent iterations, when the population becomes more concentrated, smaller steps local perturbations from the global optimal position are preferred. Consequently, the position update on theft dung beetles is defined as follows:

$$\theta_3 = 0.5 + 0.5\cos\left(\frac{\pi t}{T_{\max}}\right) \tag{16}$$

$$x_i(t+1) = \begin{cases} x_i(t) + \theta_3 \cdot S \cdot g \\ \cdot\left(|x_i(t) - X^*| + |x_i(t) - X^b|\right), \\ \qquad\qquad t \le \dfrac{T_{\max}}{2} \\ X^b + \theta_3 \cdot S \cdot g \\ \cdot\left(|x_i(t) - X^*| + |x_i(t) - X^b|\right), \\ \qquad\qquad t > \dfrac{T_{\max}}{2} \end{cases} \tag{17}$$

where $x_i(t)$ represents the position of the ith thief at the tth iteration, θ_3 is a decreasing function from 1 to 0.5, which ensures that the maximum amplitude of perturbations decreases with the number of iterations, g is a normally distributed random vector of size $1 \times D$, and S is a constant.

The SFMODBO algorithm introduces several key improvements over the MODBO, enhancing both global and local search capabilities. It begins by improving the rolling behavior through the incorporation of the Golden-SA, which utilizes sine values and the golden ratio. This enhancement is further supported by an adaptive weight mechanism that dynamically adjusts the weight during iterations, allowing for a strong global search early on and more refined local searches later, thereby preventing premature convergence. In addition, the algorithm enhances foraging using a Cauchy mutation function, which generates larger step sizes to help escape local optima and explore a wider search space. Furthermore, the improved thefting strategy in SFMODBO fully leverages current position information, taking large steps based on global and local optimal positions in the early stages to enhance exploration, and shifting to smaller, more precise steps in later iterations as the population becomes more concentrated. This strategy refines the solution, improves convergence, and prevents premature optimization, ensuring better global exploration and more effective use of position information throughout the algorithm.

The pseudo-code of the SFMODBO is shown in Algorithm 1.

Algorithm 1 Pseudo-Code of the SFMODBO

Require: the maximum number of iterations T_{max}, the population size Pop
Ensure: Pareto front P^*

```
Initialize the parameters of the SFMODBO.
Initialize the dung beetle population x_i(0)
Initialize the Pareto front P*
Initialize iteration counter t = 0
while t ≤ T_max do
    for i = 1 to Pop do
        if i is a ball-rolling dung beetle then
            Update x_i(t) by Eq. (10)
        else if i is a breeding dung beetle then
            Update x_i(t) by Eq. (13)
        else if i is a small dung beetle then
            Update x_i(t) by Eq. (15)
        else if i is a thieving dung beetle then
            Update x_i(t) by Eq. (16)
        end if
        Calculate fitness of x_i(t) for all objectives
        Update P* using non-dominated sorting
    end for
    Perform Pareto front update and remove dominated solutions
    Increment iteration counter t = t + 1
end while
return Pareto front P*
```

IV. Experiment Analysis

A. Experimental Settings

Due to the lack of standard test cases for optimizing the MOAEOSSP with three objective functions, and the difficulty

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in obtaining real-world satellite observation scheduling data, we generated reliable test data using simulation software. Specifically, the scheduling period was set from 04:00:00 to 07:00:00 on May 31, 2024, assuming all satellites are operated by Chinese authorities. The observation area covers the region within China, with latitudes ranging from 30° to 40° and longitudes from 90° to 110°. We generated scenarios containing 50 to 200 tasks, with increments of 50. Each scenario is named MOcase_50, MOcase_100, MOcase_150, and MOcase_200. The simulation software calculated all feasible time windows for satellite-to-ground station visibility, and the resulting data was processed to generate available time windows for all tasks. In addition, each task's priority was randomly assigned within the range of [1, 10]. The hardware environment for the simulation experiments included a system with an Intel[1] Core[2] i5-1340P processor with a base speed of 1.9 GHz, 16 cores, and 16-GB RAM. The system operated on a 64-bit Windows 11 environment, and the programming was conducted using MATLAB 2023b.

B. Multiobjective Evaluation Metrics

Four widely used metrics to evaluate algorithm performance are inverted generational distance (IGD) [33], generational distance (GD) [47], hypervolume (HV) [34], and Pareto front (PF) approximation error (Δ_p) [48]. Its definition is as shown in (18)–(21).

First, the IGD measures the average distance from each point on the true PF (i.e., P^*) to the nearest solution on the NDSs set (i.e., P). In the context of MOAEOSSP, smaller IGD values indicate that the algorithm's solutions are well-converged and uniformly distributed along the true PF, which is crucial for achieving the objectives of maximizing observational profit, minimizing energy consumption, and achieving load balancing

$$\text{IGD} = \frac{1}{|P^*|}\sum_{z\in P^*}\min_{x\in P} d(z,x). \tag{18}$$

1) $|P^*|$: The cardinality (number of elements) of the true PF P^*.
2) z: A solution in the true PF P^*.
3) x: A solution in the solution set P.
4) $d(z,x)$: The distance between solutions z and x, typically Euclidean distance.
5) $\min_{x\in P} d(z,x)$: The minimum distance from a solution z in P^* to the closest solution x in P.

Second, the GD calculates the average distance between the obtained solutions and the true PF, with smaller GD values indicating closer proximity to the true front. This criterion helps assess the convergence quality of the algorithm by determining how closely the obtained solutions approximate the true PF, directly impacting the optimization of the satellite scheduling objectives

$$\text{GD} = \left(\frac{1}{|P|}\sum_{x\in P}(\min_{z\in P^*} d(x,z))^p\right)^{\frac{1}{p}}. \tag{19}$$

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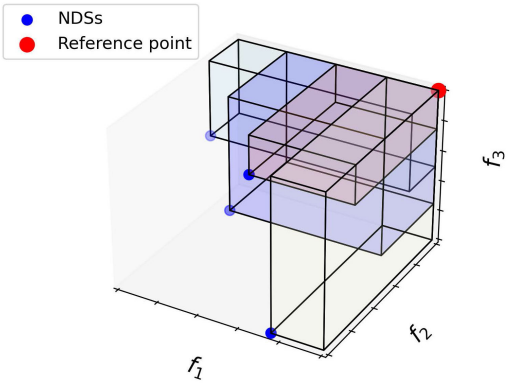


Fig. 5. Schematic of HV index calculation for three objective functions.

1) $|P|$: The cardinality (number of elements) of the solution set P.
2) x: A solution in the solution set P.
3) $\min_{z\in P^*} d(x,z)$: The minimum distance from a solution x in P to the closest solution z in P^*.
4) p: The power to which the distances are raised, typically 2 for Euclidean distance.

The HV criterion measures the volume of the objective space formed between a reference point (true PF) and the obtained solutions, as illustrated in Fig. 5. Larger HV values signify better convergence and diversity, highlighting the algorithm's ability to explore and cover a wider range of the objective space, which aligns with the goals of maximizing observational profit, minimizing energy use, and balancing load effectively

$$\text{HV} = \text{Volume}\left(\bigcup_{x\in P}[f_1(x), r_1]\times[f_2(x), r_2] \times\cdots\times[f_m(x), r_m]\right). \tag{20}$$

1) *Volume:* The volume operator, calculates the volume of the union of regions.
2) $\bigcup$: Union operator, combines multiple regions.
3) $f_i(x)$: The value of the ith objective function for solution x.
4) r_i: The reference point in the ith objective dimension.
5) m: The number of objective functions.
6) $[f_1(x), r_1]\times[f_2(x), r_2]\times\cdots\times[f_m(x), r_m]$: The hyper-rectangle defined by solution x and the reference point r.

Finally, Δ_p, which combines modified GD and IGD, measures the average Hausdorff distance between the obtained solutions and the true PF. Smaller Δ_p values are indicative of better algorithm performance, as they reflect a closer match between the solution set and the true PF in terms of both convergence and diversity, ensuring a balanced approach to the multiple objectives of MOAEOSSP

$$\Delta_p = \frac{1}{|P|\cdot\sum_{i=1}^{|P|} d_i}\cdot\sum_{i=1}^{|P|}|d_i - \bar{d}|. \tag{21}$$

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1) d_i: The distance from solution x_i to its nearest neighbor in the solution set P.
2) $\bar{d}$: The average of all d_i values.
3) $|d_i - \bar{d}|$: The absolute difference between d_i and the average distance $\bar{d}$.

C. Parameter Settings

To validate the effectiveness of the proposed algorithm, we selected SPEA2, MOEA/D, MOPSO, and the MODBO algorithm incorporating nondominated sorting as benchmarks for comparison. SPEA2 [3], [49] maintains the diversity of Pareto optimal solutions through external archives and an information distribution strategy based on dominance and density. MOEA/D [32] efficiently covers the PF by decomposing the problem into single-objective subproblems and utilizing neighborhood information. MOPSO [50], [51] employs a particle swarm optimization algorithm, combined with leader selection and mutation operations, to ensure the diversity of the solution set and convergence to the PF. By comparing these four algorithms, we comprehensively evaluated the performance of the proposed algorithm. The parameter settings for these comparison algorithms followed the recommendations in the corresponding literature, with the population size set to 200, and the external archive size to 200. In setting the appropriate number of iterations, it is crucial to consider algorithm characteristics, problem requirements, resource constraints, and solution quality demands. This consideration is particularly important when multiple algorithms are employed to solve the same problem. Consequently, we can analyze ten different iteration counts, ranging from 50 to 500, in increments of 50, as illustrated in Fig. 6. By evaluating the results using four performance metrics, we can comprehensively assess the advantages and disadvantages, enabling us to determine the optimal maximum number of iterations.

In Fig. 6, analyzing the performance of five multiobjective optimization algorithms, it is evident that selecting 300 iterations is a reasonable choice. This decision is based on the observed evaluation metrics which tend to stabilize or show diminishing returns beyond this point, indicating that the algorithms are effectively approaching the approximate PF. Notably, the algorithms exhibit significant fluctuations in performance, primarily due to the inadequate accuracy of the approximate PF and the algorithms' limitations in exploring the solution space and maintaining solution diversity. These factors can cause anomalies in evaluation metrics at specific iteration points, thereby impacting the quality of the approximated true PF. Therefore, setting 300 iterations not only helps ensure the effectiveness of the algorithms but also avoids unnecessary wastage of computational resources, achieving an optimal balance between performance and cost.

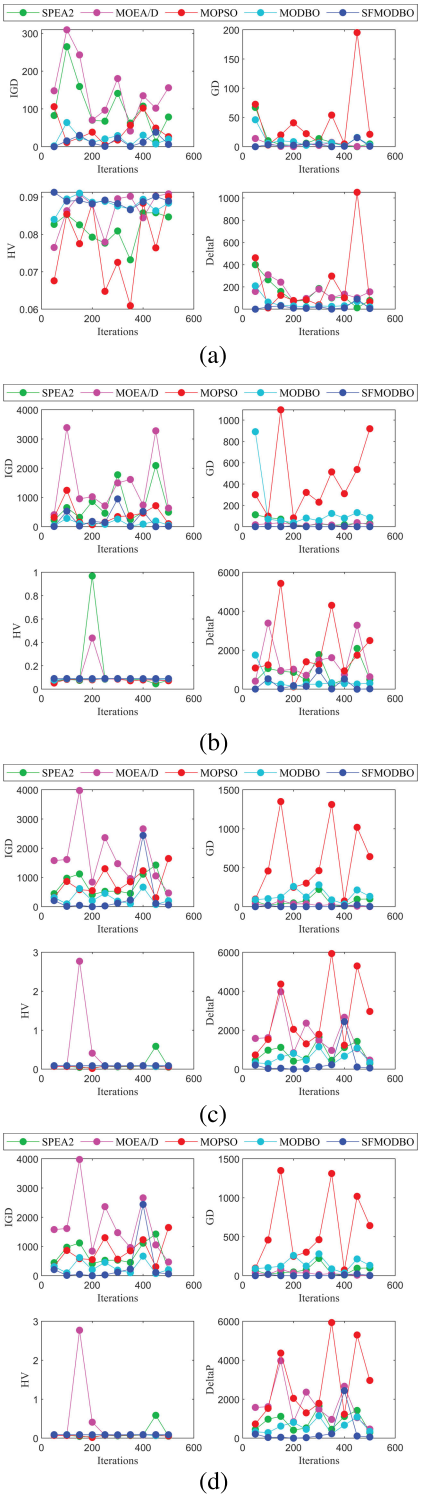


Fig. 6. Evaluation metrics by scenario and maximum iteration count. (a) MOcase_50 evaluation metrics under various maximum iterations. (b) MOcase_50 evaluation metrics under various maximum iterations. (c) MOcase_50 evaluation metrics under various maximum iterations. (d) MOcase_50 evaluation metrics under various maximum iterations.

D. Comparison Results and Analysis

Fig. 7 depicts the PF of NDS for the four scenarios. For each instance, results from random runs of SFMODBO, MODBO, and three comparison methods are retained. The x-axis represents the objective value of f_1 (maximizing observation gain), the y-axis represents f_2 (minimizing energy consumption), and f_3 (load balancing) is also considered as an objective. As shown in Fig. 7, the proposed SFMODBO not only achieves better nondominated solutions (NDSs) but also outperforms all comparison algorithms in maintaining the quality of these solutions.

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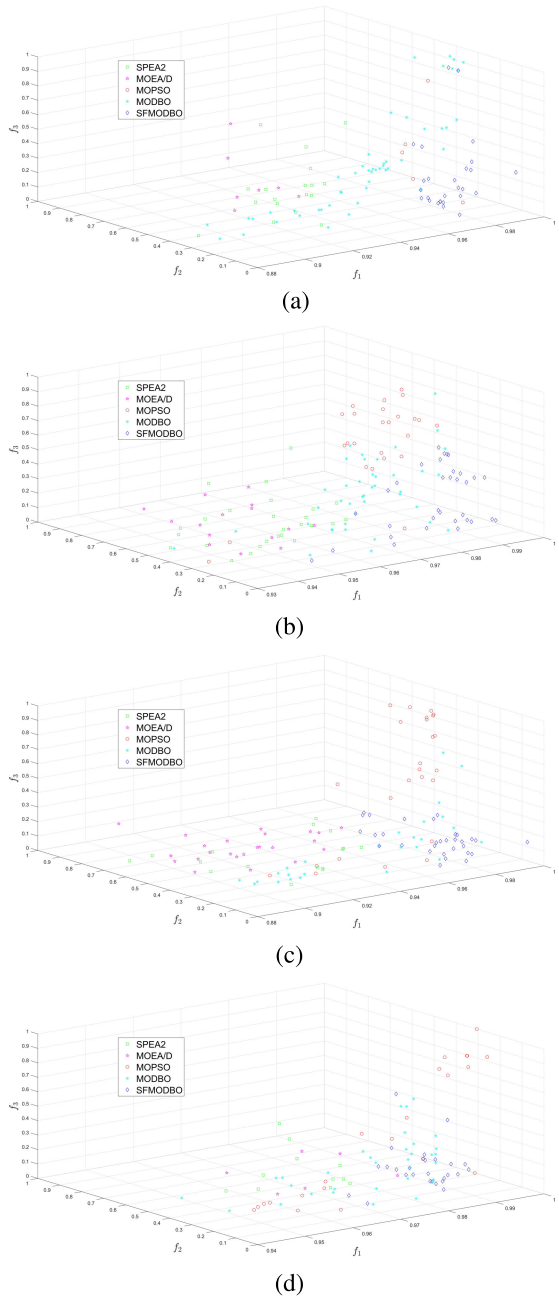


Fig. 7. SFMODBO and comparison algorithms retain NDSs in different MOAEOSSP scenarios. (a) NDSs for the MOcase_50. (b) NDSs for the MOcase_100. (c) NDSs for the MOcase_150. (d) NDSs for the MOcase_200.

From Table I, it is evident that SFMODBO performs better than most other algorithms in terms of objective function values. In particular, SFMODBO stands out as the superior algorithm among the five, especially in optimizing objective functions f_1 and f_3, where its performance is almost always the best. For objective function f_2, MODBO and SPEA2 also show commendable performance.

In practical applications, the choice of the optimal algorithm should be based on specific optimization goals and scenarios. For achieving a good balance across multiple objective functions, SFMODBO is undoubtedly the best choice. However, for optimizing specific objective functions, MODBO, SPEA2, and MOPSO also have their unique advantages.

To ensure the reliability of the experimental results, all algorithms were independently executed 30 times in each scenario. The outcomes were evaluated using four metrics: IGD, GD, HV, and Δ_p, and are displayed in box plot format. Since the true PF for each test scenario is unknown, we collected all NDS from the 30 runs of each algorithm to construct the true PF for each test scenario.

Figs. 8–11 illustrate the distribution of the four evaluation metrics for the four scenarios: MOcase_50, MOcase_100, MOcase_150, and MOcase_200. The performance of various algorithms varies significantly across different scenarios. Here is an overview of their performance as follows.

1) SFMODBO consistently outperforms other algorithms across most scenarios, excelling in both IGD and GD metrics with the lowest values and most concentrated distributions. It also achieves the highest and most stable HV values, underscoring its superior optimization performance. In terms of Δ_p, SFMODBO performs best in all scenarios except Mocase_50, where it still ranks highly. Although there is a slight drop in performance on IGD for large-scale problems, SFMODBO remains the overall top choice due to its robust performance across various metrics.
2) MODBO shows strong performance, particularly in large-scale problems, where it excels in IGD values for Mocase_150 and Mocase_200. However, its performance on GD and HV metrics is less impressive when compared to SFMODBO. MODBO consistently ranks second to SFMODBO in terms of Δ_p values, indicating good but slightly less balanced optimization capabilities.
3) SPEA2 maintains a consistent middle-ground performance across all scenarios, showing moderate results in IGD and GD metrics. It performs reasonably well on HV and Δ_p values, demonstrating stable, yet less outstanding performance. SPEA2 offers a balanced approach but does not reach the optimization levels of SFMODBO.
4) MOPSO and MOEA/D generally perform poorly, with MOEA/D particularly struggling with high IGD values and notable instability. MOPSO shows some improvement in HV values with larger scenarios but still lags significantly behind SFMODBO. Both algorithms exhibit high Δ_p values and greater variability, indicating lower stability and effectiveness compared to the other methods.

In conclusion, SFMODBO is the most effective algorithm for solving the MOAEOSSP problem, demonstrating superior optimization performance and stability across various scenarios. While MODBO shows promising results in specific large-scale contexts, it lacks the overall consistency of SFMODBO. SPEA2 provides balanced performance, offering a reasonable compromise between convergence and diversity, though it does not excel in any particular metric. In contrast, MOPSO and MOEA/D face significant limitations, with both algorithms struggling to maintain stability and effectiveness in multiobjective optimization. Each algorithm's strengths and weaknesses reflect their unique tradeoffs between diversity and convergence, illustrating the complex landscape of multiobjective optimization in the MOAEOSSP context.

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TABLE I
AVERAGE AND STANDARD DEVIATION VALUES OF THREE OBJECTIVE FUNCTIONS FOR DIFFERENT ALGORITHMS IN VARIOUS SCENARIOS

Objective	Scenarios	SPEA2		MOEA/D		MOPSO		MODBO		SFMODBO	
		Avg.	Std.	Avg.	Std.	Avg.	Std.	Avg.	Std.	Avg.	Std.
f_1	MOcase_50	0.5246	0.1553	0.3377	0.1531	0.7638	0.1597	**0.8782**	**0.0386**	0.8536	0.0795
	MOcase_100	0.3981	0.1369	0.2389	0.1567	0.8574	0.0805	0.9306	**0.0587**	**0.9407**	0.0736
	MOcase_150	0.4789	0.1406	0.3148	0.1825	0.8731	0.0894	0.8845	**0.0628**	**0.8924**	0.0720
	MOcase_200	0.5606	0.1123	0.2304	0.1012	0.8800	**0.0231**	0.9010	**0.0248**	**0.9359**	0.0385
f_2	MOcase_50	**0.5137**	0.2311	0.8454	0.1184	0.7118	0.0916	0.6815	**0.0661**	0.6682	0.1160
	MOcase_100	**0.5308**	0.2276	0.8081	0.1761	0.6655	0.1045	0.6151	**0.0749**	0.5779	0.0951
	MOcase_150	0.5568	0.1024	0.6191	0.3041	0.5948	0.1049	**0.4508**	**0.0680**	0.4988	0.0941
	MOcase_200	0.6602	0.0902	0.6032	0.3246	0.6569	0.1322	0.5287	**0.0682**	**0.5082**	0.1013
f_3	MOcase_50	0.3765	0.1730	0.4572	0.2348	0.6819	0.2486	0.2302	0.1848	**0.0330**	**0.1396**
	MOcase_100	0.1995	0.0926	0.4200	0.1022	0.5694	0.2970	0.1171	0.0713	**0.0441**	**0.0418**
	MOcase_150	0.1828	0.0357	0.3407	0.0878	0.5154	0.3314	0.0945	0.0717	**0.0054**	**0.0128**
	MOcase_200	0.1841	0.0251	0.3099	0.1091	0.6045	0.3056	0.1093	0.0566	**0.0024**	**0.0042**

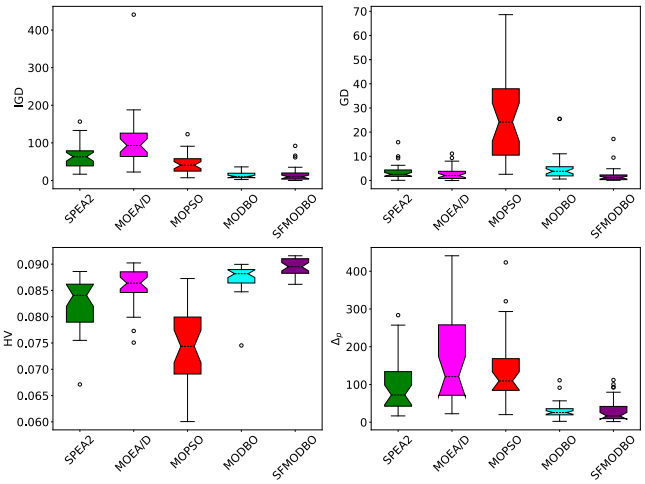


Fig. 8. Distribution of the four evaluation metrics for the MOcase_50 scenario.

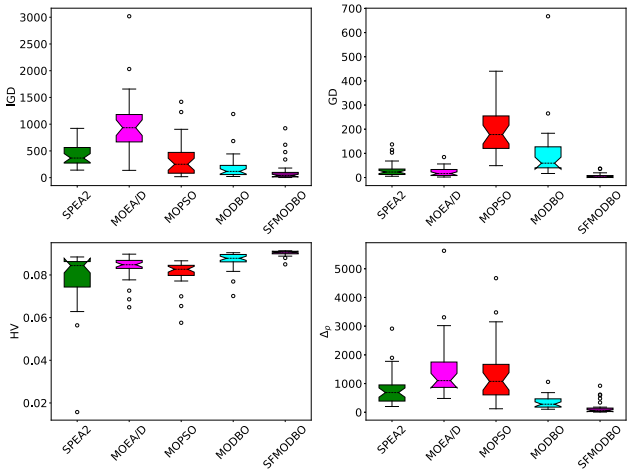


Fig. 9. Distribution of the four evaluation metrics for the MOcase_100 scenario.

By conducting 30 runs of five different algorithms in the MOcase_150 scenario, the execution time and memory usage of each algorithm were evaluated to analyze their efficiency and stability. This testing aims to comprehensively understand the performance of each algorithm, with a particular focus on time efficiency and resource utilization. Fig. 12 presents the kernel density estimation (KDE) and cumulative distribution function (cdf) plots of the execution times for the algorithms. Fig. 13 shows a violin plot of their memory usage.

The advantages of algorithms with shorter execution times are evident from the KDE and cdf plots, where MOPSO and MODBO have relatively shorter execution times, with KDE peaks concentrated within 10 s, indicating a high probability of task completion in this time frame, suggesting a faster response speeds. Correspondingly, their cdf curves rapidly rise to 1, showing that the majority of tasks are completed in a very short time. In contrast, SPEA2 and MOEA/D exhibit longer execution times, with SPEA2 peaking near 120 s, and its cdf curve only reaching 100% at 120 s, suggesting these algorithms may be more suitable for complex scenarios where longer execution times are acceptable, potentially offering better optimization or stability. SFMODBO falls in between, with the KDE showing its execution time is mainly concentrated around 10 s, while the cdf plot suggests that most tasks are completed in a short time, although a small number may require longer execution times.

From Fig. 13, it can be seen that SFMODBO exhibits relatively low and stable memory usage compared to the other algorithms. While MOPSO and MODBO also demonstrate low memory usage, their distributions are broader, suggesting some variability. SPEA2 shows the highest memory

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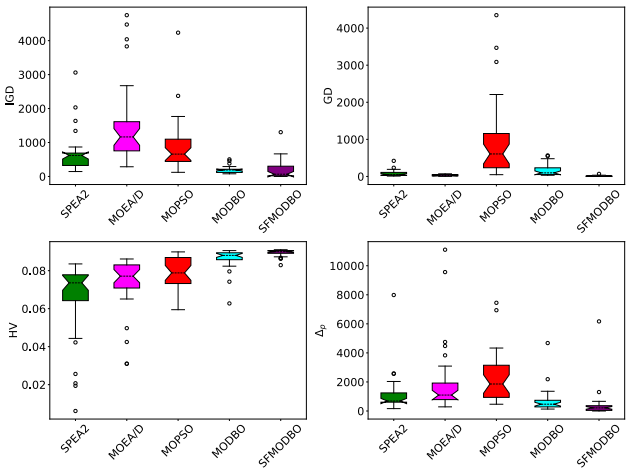


Fig. 10. Distribution of the four evaluation metrics for the MOcase_150 scenario.

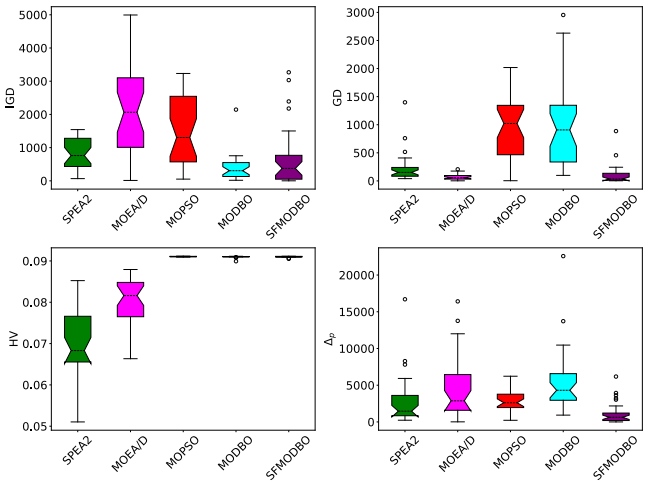


Fig. 11. Distribution of the four evaluation metrics for the MOcase_200 scenario.

usage but remains more stable than MOEA/D, which has the widest range and the most pronounced fluctuations. Overall, SFMODBO stands out as an optimal choice for solving the MOAEOSSP problem, balancing relatively low memory consumption with stable performance.

To summarize, this section demonstrates that the proposed SFMODBO algorithm, when compared to several classical algorithms, exhibits superior convergence, distribution, and expansion capabilities, making it a strong candidate for addressing the MOAEOSSP problem. Through comparisons of the NDSs and four evaluation metrics across various test scenarios, SFMODBO consistently shows good performance. Notably, SFMODBO and MODBO excel in IGD, with SFMODBO further distinguishes itself with significant advantages in GD, HV, and Δ_p metrics. Although SFMODBO is not the top performer in terms of runtime and memory usage, it offers greater overall stability. These results highlight SFMODBO's robustness and effectiveness, demonstrating its practical significance for solving the MOAEOSSP problem.

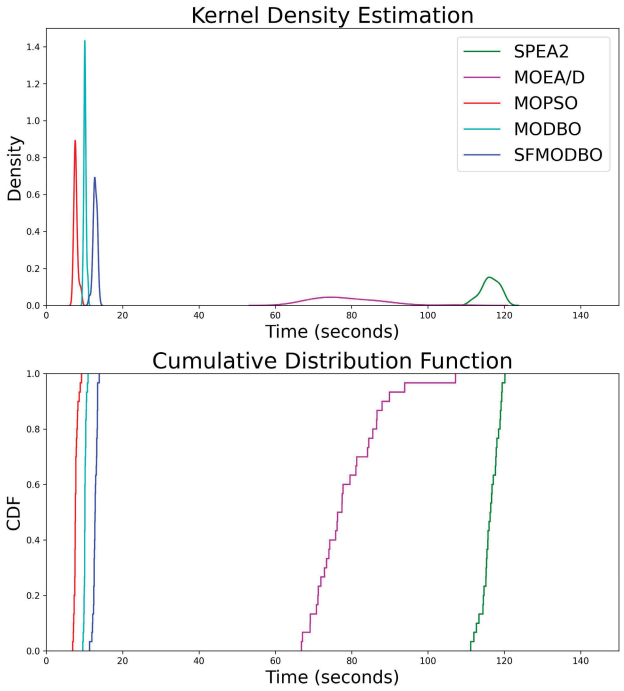


Fig. 12. KDE and cdf of execution times for different algorithms in the MOcase_150 scenario.

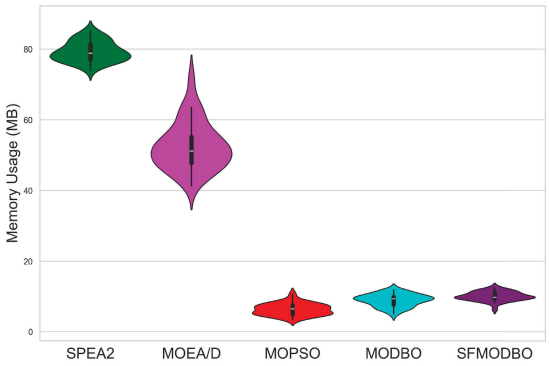


Fig. 13. Violin plot of memory usage for different algorithms in the MOcase_150 scenario.

V. Conclusion and Future Work

This study successfully proposes and validates a multiobjective optimization mathematical model designed to solve the MOAEOSSP, filling a crucial gap in the existing literature that has predominantly focused on bi-objective functions. By innovatively integrating multiple strategies, we developed the novel optimization algorithm SFMODBO. Through extensive simulation experiments, SFMODBO demonstrated a significant ability to balance profit, energy consumption, and load balancing, surpassing multiple competing algorithms across various evaluation metrics. These results underscore SFMODBO's potential to effectively address the complexities of the MOAEOSSP.

However, we acknowledge that the diversity and number of simulation scenarios in this study may limit the generalizability of the results. Future research should broaden the scope of testing scenarios to further evaluate the robustness of SFMODBO in more complex environments. Moreover,

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further optimization of the algorithm is necessary to enhance its performance. Another promising direction for future research involves applying SFMODBO to other multiobjective optimization problems across different domains, where its application could lead to more efficient solutions across different operational scenarios.

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